Audit Log Fields and Why Each Is Useful

Each line in /var/log/audit/audit.log is a record made of key=value fields that show who did what, when, from where, and whether it succeeded.

Fields and their use

Field	Meaning	Why it is useful
`type`	Kind of record (SYSCALL, EXECVE, PATH, CWD, USER_CMD, USER_AUTH, USER_LOGIN, BPF, CRED_ACQ)	Lets you filter by category, such as only logins or only commands.
`msg=audit(epoch:serial)`	Unix timestamp and unique event serial number	Gives the exact time; records sharing a serial belong to one event, so SYSCALL, EXECVE and PATH lines can be correlated.
`arch`	System architecture	Needed to interpret syscall numbers correctly.
`syscall`	System call number (59 = execve on x86_64)	Shows what the kernel was asked to do: execute, open, delete and so on.
`success` / `res`	Whether the action succeeded or failed	Separates real changes from failed attempts, which exposes brute-force or denied access.
`exit`	Return code of the syscall	Gives the exact result or error (-13 = permission denied).
`pid`	Process ID	Identifies the process that performed the action.
`ppid`	Parent process ID	Shows what launched the process, which helps trace an attack chain.
`uid`	User ID the process runs as	Shows who ran it (0 = root).
`gid`	Group ID of the process	Shows group privileges for access-control reviews.

Field	Meaning	Why it is useful
`auid`	Audit (login) user ID	The original logged-in user, even after sudo or su; the key field for accountability. 4294967295 means unset (system or daemon).
`ses`	Session ID	Groups all activity from one login session.
`euid`, `egid`, `suid`, `fsuid`	Effective, saved and filesystem IDs	Detects privilege escalation, such as a normal user running as root through a setuid binary.
`subj`	SELinux or AppArmor security context (`unconfined` in this log)	Shows which security policy applied to the process.
`exe`	Full path of the executable (`/usr/bin/sudo`)	Shows exactly which program ran, so unusual binaries stand out.
`comm`	Short command name	Quick, readable name of the process.
`cmd` (USER_CMD), `a0`, `a1` (EXECVE)	Command line and arguments	Shows exactly what was typed and run.
`cwd`	Current working directory	Shows where the command was run from.
`name`, `inode`, `mode` (PATH)	File path, inode and permissions	Shows which file was accessed or changed.
`terminal` / `tty`	Terminal used (`/dev/pts/0`, `cron`)	Shows whether the action came from an interactive shell, remote session or scheduled job.
`acct`	Account name involved	Shows which account was authenticated or used (root via sudo).
`hostname`, `addr`	Remote host and IP address	Shows where a login came from (`?` means local).
`op`	PAM operation (`PAM:session_open`, `PAM:setcred`)	Shows the stage of authentication or session handling.
`grantors`	PAM modules that approved the action	Shows which authentication rules allowed access.

Field	Meaning	Why it is useful
`key`	Custom tag from an audit rule (`-k nano`)	Lets you find your own rule's events quickly with `ausearch -k`.
`prog-id`, `op=LOAD/UNLOAD` (BPF)	eBPF program ID and action	Tracks kernel-level programs being loaded, which matters for security monitoring.
`unit` (SERVICE_START/STOP)	systemd service name	Tracks services starting and stopping.

Reading the log

The `cmd` value in a USER_CMD record is hex-encoded (for example `cmd=6175736561726368D6B206E616E6F`).